Notes: Autor and Scarborough (QJE, 2008)

Does Job Testing Harm Minority Workers? Evidence from Retail Establishments

Contents

1 Big picture

- **Question.** Does computerized job testing improve worker selection at the cost of reducing minority hiring, or can testing raise productivity without creating a meaningful equality–efficiency trade-off?
- **Setting.** A national retail firm introduced a computerized applicant assessment across 1,363 stores over roughly one year. This staggered rollout creates variation in whether applicants and hires were screened by the old informal process or by the new formal test.
- **Core concern.** Standardized tests often show lower average scores for minority groups. A common policy and legal concern is that tests improve productivity but reduce minority employment opportunities.
- **Conceptual distinction.** The paper separates two effects of testing: *screening precision* and *belief change*. If both interviews and tests are unbiased, testing can improve precision without reducing minority hiring. If the test adds majority-favoring information relative to the old interview, minority hiring can decline.
- **Main empirical result.** Testing raises worker productivity, measured primarily by job tenure. Mean and median job spells rise meaningfully after testing is adopted.
- **Minority hiring result.** Even though black and Hispanic applicants score lower on the test on average, the introduction of testing has no statistically meaningful negative effect on minority hiring shares.
- **Race-specific productivity result.** Productivity gains from testing are large for white and black hires and not statistically different across groups. Hispanic estimates are noisier but do not imply a clear equality–efficiency trade-off.
- **Mechanism.** The old informal screen was already using information correlated with worker productivity. The formal test appears to increase precision rather than add new negative information about minorities.
- **Policy implication.** Disparate test-score distributions do not mechanically imply disparate employment impacts. Whether testing harms minority workers depends on the relationship between the test, the old screening method, and true productivity.

2 Incremental contribution

- **Conceptual contribution to discrimination and testing.** The paper moves beyond the simple claim that lower minority test scores imply reduced minority employment. It shows that the effect of testing depends on whether the test is biased relative to the existing screening method.
- **Separation of precision from bias.** Prior discussions often bundle testing together with adverse impact. Autor and Scarborough separate *precision improvement* from *belief revision*. This distinction explains why testing can improve productivity without changing minority hiring.
- **Empirical contribution using real personnel data.** The paper studies almost 190,000 applications and 33,924 hires at a national retail firm, rather than relying on laboratory tests, military settings, or small personnel samples.
- **Quasi-experimental rollout.** The staggered adoption of testing across stores permits within-store and over-time comparisons of hiring and worker tenure.
- **Productivity plus hiring.** The paper jointly measures the effect of testing on both productivity and demographic composition. Many studies only observe test scores or hiring; this one observes applicant scores, actual hires, and subsequent job durations.
- **Contribution to Title VII and disparate-impact debates.** The paper provides a framework for evaluating when job tests generate a trade-off between efficiency and equality, relevant to employment testing regulation.

- **Simulation contribution.** The paper calibrates simulations using the firm's actual applicant and hire distributions to show which combinations of test bias and interview bias can match the observed data.
- **Broader methodological contribution.** The paper illustrates how a firm-level technology adoption can be used to distinguish selection quality effects from demographic composition effects.

3 Data and measurement

3.1 Firm setting and rollout

- The employer is a large national retail firm with 1,363 sites.
- Prior to testing, stores used an informal screening process, centered on applications and interviews.
- During the rollout, stores adopted a computerized assessment system that scored applicants and categorized them into broad test-score groups.
- The rollout timing differs across sites, allowing comparisons of workers hired before and after testing within the same firm.

Interpretation: The study is not comparing different employers with different HR policies. It studies a single firm changing its selection technology, which reduces cross-firm confounding.

3.2 Applicant and hire data

- The application sample contains 189,067 applicants and 16,925 hires at treatment sites.
- The full hire sample contains 33,924 hired workers.
- The paper observes race/ethnicity, gender, test scores, hire status, store, date, and job spell duration.
- Job spell duration is the main productivity measure, motivated by the high cost of turnover in low-wage retail jobs.

Interpretation: Turnover is a practical productivity proxy: longer tenure means less hiring, training, and vacancy cost. It is not a complete measure of effort or quality, so the paper uses robustness checks and interprets it carefully.

3.3 Test score categories

- The standardized test score is normalized to zero mean and unit variance in the applicant sample.
- The test-score distribution is divided into broad categories: red, yellow, and green.
- The red category has extremely low hiring rates, while green applicants account for most hires.
- Minority applicants are overrepresented in lower test-score categories, but the difference between applicant-score distributions does not automatically translate into differences among hires.

Interpretation: Hiring decisions are highly selective. Even if applicant distributions differ by race, the distribution of hired workers can be much closer across race groups because the firm hires mainly from the upper part of each group's score distribution.

3.4 Treatment and outcomes

- The primary treatment indicator is whether the store had adopted computerized testing at the time of hire.
- Main productivity outcomes are job spell duration, log job spell duration, and completed employment spell length.
- Main demographic outcomes are the race and ethnicity composition of hires and minority hiring odds.
- The paper also studies site-level minority shares before and after adoption of testing.

Interpretation: The treatment is a store-level screening technology. Individual receipt of the test can be instrumented by site adoption because not every observed hire necessarily has an individually recorded test result.

4 Models and empirical specifications

4.1 Conceptual model: productivity, interviews, and tests

Let y denote true applicant productivity. The pre-existing interview signal and the job-test signal can be written schematically as

$$\eta = y + b_\eta(g) + \varepsilon_\eta, \qquad s = y + b_s(g) + \varepsilon_s,$$

where g is group membership, $b_\eta(g)$ captures possible interview bias, $b_s(g)$ captures possible test bias, and the error terms capture signal noise.

Interpretation: If the job test reduces noise relative to interviews but is not more biased by group, testing raises precision without necessarily reducing minority hiring. If the test is more majority-favoring than interviews, it can lower minority hiring.

4.2 Productivity regression

The basic empirical specification for productivity is of the form

$$Y_{ijt} = \alpha + \beta Test_{jt} + X_i'\gamma + \theta_t + \phi_j + \varepsilon_{ijt},$$

where Y_{ijt} is the job spell of worker i hired at store j in month t, $Test_{jt}$ is the indicator that store j has adopted testing, X_i includes demographic controls, θ_t are month-year effects, and ϕ_j are site effects.

Interpretation: The site fixed effects compare the same store before and after testing. Time effects remove aggregate seasonal or macro shocks. The coefficient β captures the change in tenure associated with the new testing regime.

4.3 Race-specific productivity regression

To test whether productivity gains differ by group, the paper estimates models with interactions:

$$Y_{ijt} = \alpha + \beta_W(White_i \times Test_{jt}) + \beta_B(Black_i \times Test_{jt}) + \beta_H(Hispanic_i \times Test_{jt}) + X_i'\gamma + \theta_t + \phi_j + \varepsilon_{ijt}.$$

Interpretation: If testing improves productivity for all groups similarly, the equality–efficiency trade-off is weak. If gains occur only by substituting away from minority workers or by selecting only high-scoring majority workers, group-specific gains would look different.

4.4 Instrumental variables specification

The paper also uses site adoption as an instrument for whether an individual hire was tested. The first stage is whether the worker received the test, instrumented by whether the store had begun testing.

Interpretation: The IV specification addresses imperfections in individual test recording and captures the causal effect of being selected under the new regime rather than merely the correlation of observed test status with worker characteristics.

4.5 Minority hiring model

For the probability that a hire is from a minority group, the paper estimates conditional logit and linear probability models. A simplified version is

$$Pr(x_2 \mid Hire = 1) = F(\xi T_i + X_i'\beta + \theta_t + \phi_j),$$

where x_2 indicates that the hire is from the group of interest and T_i indicates testing.

Interpretation: Because the paper observes hires but not all application flows by race at all sites before testing, the key assumption is that within-store changes in minority application rates are not driving the results. The store fixed effects absorb time-invariant application mix.

4.6 Simulation framework

The simulation compares six scenarios: two assumptions about test bias and three assumptions about interview bias. The model uses the empirical distribution of applicants, hiring rates, and test scores to predict changes in tenure and hiring composition under each bias scenario.

Interpretation: The simulations are a bridge between the conceptual model and the data. They ask which bias scenario could reproduce the observed facts: large productivity gains, no minority hiring decline, and similar black-white tenure gains.

5 Identification and empirical design

5.1 Core identification problem

The key empirical problem is that job testing is not randomly assigned to workers. Stores adopt the system over time, and stores may differ in management quality, local labor markets, and hiring practices. The paper addresses this by comparing stores before and after adoption, controlling for site fixed effects, time effects, state trends, and worker characteristics.

5.2 Why the rollout helps

- Adoption occurs across many sites, giving many before-after comparisons.
- Month-year controls absorb common shocks to retail labor markets.
- Site fixed effects absorb permanent differences across stores.
- State trends absorb local labor-market and policy changes correlated with adoption timing.

Interpretation: The design is not a randomized controlled trial, but it uses within-store changes and a large multi-site rollout to limit confounding.

5.3 Testing equality-efficiency trade-offs

A trade-off would imply that productivity gains come with reduced minority hiring or lower relative opportunities for minority applicants. The paper evaluates this using three types of evidence:

- changes in job spell duration after testing,
- changes in minority hiring shares after testing,
- simulations of different test-bias and interview-bias scenarios.

Interpretation: The strongest interpretation comes from the joint pattern: testing raises tenure substantially, but minority hiring does not fall measurably.

5.4 Threats and limitations

- Job tenure is not a perfect productivity measure.
- The paper does not observe pre-testing application flows by race with the same completeness as hires.
- Testing may change who applies to the firm, especially if electronic kiosks deter some applicants.
- The results come from one national retail employer and may not generalize to all jobs or tests.
- The paper cannot directly observe true worker productivity or the content of interviews.

Interpretation: These limitations matter, but the paper's combined evidence narrows the plausible explanations for its main finding.

6 Section-by-section study guide

- **Introduction.** Explains the perceived trade-off between productivity and minority hiring and motivates the conceptual distinction between precision and bias.
- **Conceptual framework.** Shows why testing need not harm minorities if the old screen and test are both unbiased.
- **Data section.** Describes the retail firm, store rollout, applicant scores, hire data, and job spell measures.
- **Productivity section.** Estimates the effect of testing on tenure and shows that productivity gains are large.
- **Minority hiring section.** Tests whether testing reduces minority hiring and finds little evidence that it does.
- **Simulation section.** Compares observed results to alternative assumptions about test and interview bias.
- **Conclusion.** Argues that employment testing can improve screening precision without creating a measurable disparate impact.

7 Tables and figures

All figures and tables below are cropped directly from the paper. Close visuals are placed on the same row. When a row contains only one visual, the width is set to 0.6 of the text width.

7.1 Table I: Race and Gender Characteristics of Tested and Nontested Hires + Table II: Test Scores and Hire Rates by Race and Sex for Tested Applicant Subsample

Read:

- Table I shows that 25% of full-sample hires are tested hires, while 75% are nontested hires.
- Among tested hires, black and Hispanic shares are not lower than among nontested hires; if anything, black and Hispanic shares are slightly higher in the tested sample.
- Tested hires have slightly longer median tenure than nontested hires: 107 days versus 96 days overall.
- Table II shows that black and Hispanic applicants score lower on the test on average than white applicants.
- However, almost all hires come from the green category. The red-category hiring rate is near zero.

Detailed interpretation: Table I is important because it gives the first descriptive check on the alleged disparate impact. If testing mechanically harmed minority employment, one would expect minority shares among tested hires to fall sharply. Table II explains why the concern is still real: the applicant test-score distributions differ by race. The paper's puzzle is that lower average scores do not translate into reduced minority hiring once the firm selects from the top part of each distribution.

Economic message: Descriptive patterns already point to the paper's central result: testing changes selection quality more than group composition.

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TABLE I
RACE AND GENDER CHARACTERISTICS OF TESTED AND NONTESTED HIRES

A. Frequencies

	Full sample		Nontested hires		Tested hires	
	Frequency	% of total	Frequency	% of total	Frequency	% of total
All	33,924	100	25,561	75	8,363	25
White	23,560	69.5	18,057	70.6	5,503	65.8
Black	6,262	18.5	4,591	18.0	1,671	20.0
Hispanic	4,102	12.1	2,913	11.4	1,189	14.2
Male	17,444	51.4	13,008	50.9	4,436	53.0
Female	16,480	48.6	12,553	49.1	3,927	47.0

B. Employment spell duration (days)

	Full sample		Nontested hires		Tested hires	
	Median	Mean	Median	Mean	Median	Mean
All	99	173.7	96	173.3	107	174.8
	[97, 100]	(1.9)	[94, 98]	(2.1)	[104, 111]	(2.9)
White	106	184.0	102	183.0	115	187.1
	[103, 108]	(2.1)	[100, 105]	(2.3)	[112, 119]	(3.6)
Black	77	140.1	74	138.1	87	145.7
	[75, 80]	(3.0)	[71, 77.4]	(3.5)	[81.9, 92]	(4.8)
Hispanic	98	166.4	98	169.3	99	159.5
	[93, 103]	(4.6)	[92, 104]	(5.4)	[90, 106]	(6.4)

Sample includes workers hired between January 1999 and May 2000. Mean tenures include only completed spells (98% spells completed). Median tenures include complete and incomplete spells. Standard errors in parentheses account for correlation between observations from the same site (1,363 sites total). 95 percent confidence intervals for medians are given in brackets.

Table I: race/gender composition and tenure

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TABLE II
TEST SCORES AND HIRE RATES BY RACE AND SEX FOR TESTED APPLICANT SUBSAMPLE

A. Test scores of applicants (n = 189,067)

			Percentage in each category		
	Mean	SD	Quartile 1: "red"	Quartile 2: "yellow"	Quartiles 3 and 4: "green"
All	0.000	1.000	23.2	24.8	52.0
White	0.064	0.996	20.9	24.5	54.6
Black	−0.125	1.009	27.8	25.2	47.1
Hispanic	−0.056	0.982	24.9	25.6	49.6
Male	0.019	0.955	24.4	24.3	51.3
Female	−0.014	1.033	21.6	25.5	52.9

B. Test scores of hires (n = 16,925)

			Percentage in each category		
	Mean	SD	Quartile 1: "red"	Quartile 2: "yellow"	Quartiles 3 and 4: "green"
All	0.707	0.772	0.18	16.1	83.8
White	0.720	0.772	0.14	15.7	84.2
Black	0.667	0.777	0.39	16.4	83.2
Hispanic	0.695	0.768	0.13	17.3	82.6
Male	0.749	0.750	0.23	14.9	84.8
Female	0.657	0.788	0.13	17.4	82.5

C. Hire rates by applicant group

By Race and Sex			By Test Score Decile		
Race/Sex	% Hired	Obs	Decile	% Hired	Obs
			1	0.07	19,473
All	8.95	189,067	2	0.06	20,038
			3	3.96	18,803
White	10.16	113,354	4	5.65	18,774
Black	7.17	43,314	5	7.97	19,126
Hispanic	7.12	32,399	6	10.99	18,264
			7	11.71	18,814
			8	13.76	18,029
Male	8.59	106,948	9	16.14	19,491
Female	9.42	82,119	10	20.43	18,255

N = 189,067 applicants and 16,925 hires at 1,363 sites. Sample includes all applicants and hires between August 2000 and May 2001 at sites used in treatment sample. Test score sample is standardized with zero mean and unit variance.

Table II: test scores and hire rates

7.2 Figure I: Conditional Probability of Hire as a Function of Test Score + Figure II: Test Score Distributions for Applicants and Hires

Read:

- Figure I shows a steep relationship between test scores and hiring probability.
- Applicants in the lower tail are almost never hired, while hiring probability rises sharply at high scores.
- Figure II shows that the score distributions of white, black, and Hispanic applicants differ meaningfully.
- Among hires, however, the racial score distributions look much more similar than among applicants.

Detailed interpretation: Figure I shows that managers use the test score strongly in hiring decisions. Figure II shows that the firm's high hiring threshold compresses differences among hires. These visuals together explain why lower average minority scores do not necessarily imply lower minority hiring: selection is concentrated in the upper tail, and each group has applicants in that upper tail.

Economic message: Testing can be consequential for productivity while still having limited effects on minority hiring if it mainly ranks applicants within each group.

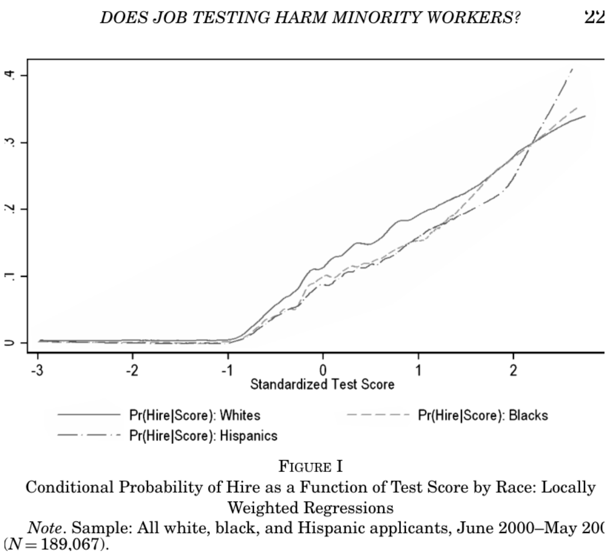

Figure I: conditional hiring probability by test score

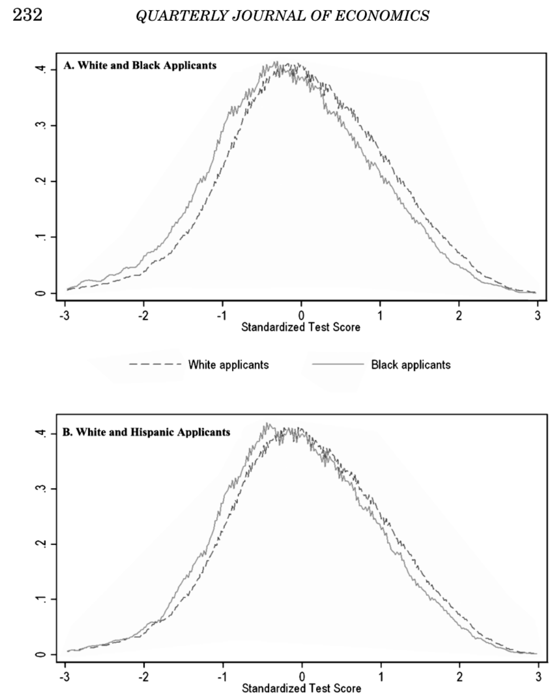

Figure II: score distributions by race

7.3 Table III: The Relationship Between Applicant Characteristics and Test Scores + Figure III: Simulated Impact of Job Testing on Hiring and Productivity

Read:

- Table III confirms that black and Hispanic applicants have lower standardized test scores even after controls.
- Male applicants also have somewhat lower scores in these specifications.
- Adding site effects and local socioeconomic controls reduces some gaps but does not eliminate them.
- Figure III shows simulated trade-offs under different assumptions about testing and interview bias.
- If tests and interviews are unbiased, productivity gains can occur without a large minority hiring penalty.

Detailed interpretation: Table III documents the raw material for potential disparate impact: group-level score differences. Figure III shows that the presence or absence of a trade-off depends on relative bias. If the prior informal interview already contains the same group-related information as the test, a more precise test does not necessarily change group hiring rates much. If the test is more majority-favoring than interviews, the trade-off becomes stronger.

Economic message: Lower test scores for minority applicants are not enough to prove harm. The relevant comparison is between the test and the pre-existing screening technology.

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TABLE III
THE RELATIONSHIP BETWEEN APPLICANT CHARACTERISTICS AND TEST SCORES
(DEPENDENT VARIABLE: STANDARDIZED TEST SCORE)

	(1)	(2)	(3)	(4)	(5)
Black	−0.192	−0.183	−0.125	−0.113	−0.113
	(0.008)	(0.007)	(0.008)	(0.008)	(0.008)
Hispanic	−0.121	−0.148	−0.100	−0.093	−0.093
	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)
Male	−0.044	−0.045	−0.052	−0.053	−0.053
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Median income in applicant's ZIP code				0.066	0.062
				(0.015)	(0.016)
Percent nonwhite in applicant's ZIP code				−0.071	−0.071
				(0.023)	(0.023)
State effects	No	Yes	No	No	No
1,363 site effects	No	No	Yes	Yes	Yes
State trends	No	No	No	No	Yes
R^2	0.0070	0.0113	0.0265	0.0269	0.0277
Obs			189,067		

Robust standard errors in parentheses account for correlation between observations from the same site (1,363 sites). Sample includes all applications from August 2000 through May 2001 at sites in treatment sample. All models include controls for the year-month of application and an "other" race dummy variable to account for 25,621 applicants with other or unidentified race. Income and fraction nonwhite for stores and applicants are calculated using store ZIP codes merged to 2000 Census SF1 and SF3 files.

Table III: applicant characteristics and scores

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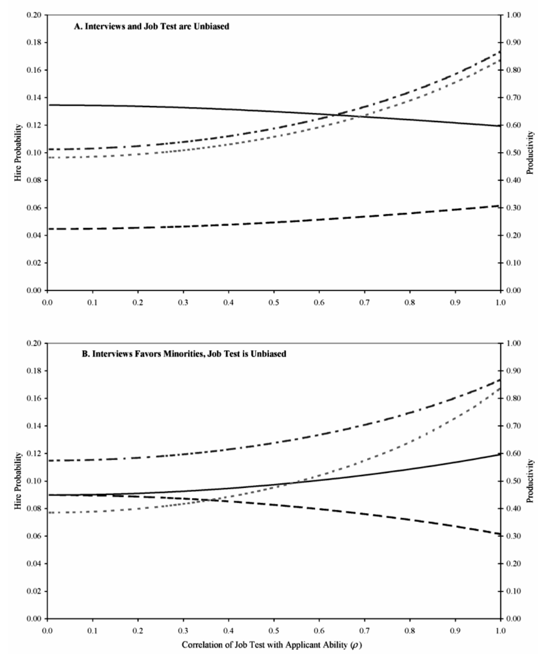

Figure III: simulated effects of testing

7.4 Table IV: OLS and IV Estimates of the Effect of Job Testing on Job Spell Duration of Hires + Figure IV: Distribution of Job Spell Duration

Read:

- Table IV estimates positive effects of testing on job spell duration.
- OLS estimates become larger when site effects and state trends are included.
- 2SLS estimates using site adoption as an instrument are also positive.
- Figure IV shows that tested hires have a distribution of completed spell lengths shifted toward longer employment spells.
- The distributional evidence supports the regression evidence: productivity improvements are not driven by a tiny number of outliers.

Detailed interpretation: Table IV is the main productivity result. The dependent variable is completed employment spell length. The effect is economically meaningful because turnover is high in retail and average spells are short. Figure IV confirms the shape of the effect: testing appears to reduce very short spells and improve retention throughout the distribution.

Economic message: Testing raised hiring productivity by selecting workers who stayed longer.

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TABLE IV
OLS AND IV ESTIMATES OF THE EFFECT OF JOB TESTING ON THE JOB SPELL DURATION OF HIRES
(DEPENDENT VARIABLE: LENGTH OF COMPLETED EMPLOYMENT SPELL IN DAYS)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	OLS estimates						2SLS estimates			
Employment test			8.9	18.4	18.4	21.8	6.3	14.9	14.8	18.1
			(4.5)	(4.0)	(4.0)	(4.3)	(5.1)	(4.6)	(4.6)	(5.0)
Black	−43.5	−25.9			−25.9	−25.8			−25.9	−25.8
	(3.2)	(3.5)			(3.5)	(3.5)			(3.5)	(3.5)
Hispanic	−17.5	−11.8			−11.8	−11.7			−11.8	−11.7
	(4.4)	(4.1)			(4.1)	(4.1)			(4.1)	(4.1)
Male	−4.2	−2.0			−2.0	−1.9			−2.0	−1.9
	(2.4)	(2.4)			(2.4)	(2.4)			(2.4)	(2.4)
Site effects	No	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
State trends	No	No	No	No	No	Yes	No	No	No	Yes
R^2	0.0112	0.1089	0.0049	0.1079	0.1094	0.1116				

$N = 33{,}266$. Robust standard errors in parentheses account for correlation between observations from the same site hired under each screening method (testing or no testing). All models include controls for month-year of hire. Sample includes workers hired January 1999 through May 2000 at 1,363 sites. Instrument for worker receiving employment test in columns (7)–(10) is an indicator variable equal to one if site has begun testing.

Table IV: testing and job spell duration

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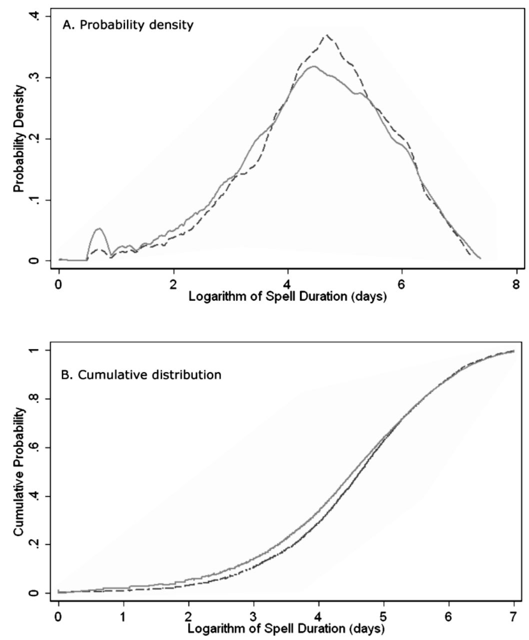

Figure IV: job spell distribution

7.5 Table V: The Relationship Between Site-Level Mean Test Scores and Job Spell Duration of Hires + Table VI: OLS and IV Estimates of the Effect of Job Testing on the Job Spell Duration of Hires by Race

Read:

- Table V links site-level mean test scores to worker tenure.
- Higher mean test scores of applicants and hires are associated with longer job spells.
- Table VI estimates race-specific effects of testing on tenure.
- White and black workers experience very similar tenure gains under testing.
- Hispanic estimates are smaller and noisier but do not statistically reject equality of effects across groups.

Detailed interpretation: Table V validates the test as a productivity-relevant screen. Table VI is crucial for equality-efficiency interpretation: if the productivity gains came only from replacing minority workers with majority workers, gains would be concentrated in one group. Instead, black and white hires both show large tenure improvements.

Economic message: The productivity benefit of testing applies broadly across groups, weakening the claim that testing improves efficiency by excluding minorities.

TABLE V

THE RELATIONSHIP BETWEEN SITE-LEVEL MEAN TEST SCORES AND JOB SPELL DURATION OF HIRES

(DEPENDENT VARIABLE: LENGTH OF EMPLOYMENT SPELL IN DAYS)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Nontested hires (OLS)			Tested hires (OLS)			Tested hires (2SLS)		
Mean test score of applicants	39.2	40.3	36.5	36.4	36.2	40.9			
	(11.9)	(12.1)	(13.6)	(18.3)	(18.3)	(19.7)			
Mean test score of hires							57.5	57.0	53.9
							(25.8)	(25.5)	(23.7)
Log median income in store ZIP code			−12.3			−23.7			−18.4
			(7.1)			(11.2)			(11.5)
Share nonwhite in store ZIP code			−19.4			−12.8			−21.5
			(11.2)			(16.2)			(15.3)
Black	−37.2	−36.6	−34.8	−34.2	−33.2	−33.8	−35.8	−34.9	−33.3
	(4.0)	(4.0)	(4.1)	(6.1)	(6.0)	(6.3)	(6.0)	(5.9)	(6.2)
Hispanic	−9.9	−9.7	−8.2	−23.2	−22.9	−24.1	−25.7	−25.5	−24.7
	(5.5)	(5.5)	(5.3)	(7.0)	(7.0)	(7.1)	(7.1)	(7.1)	(7.2)
Male	−5.4	−5.5	−5.3	0.0	−0.7	−0.2	0.3	−0.5	−0.3
	(2.8)	(2.8)	(2.8)	(4.8)	(4.8)	(4.9)	(4.8)	(4.8)	(4.8)
State effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State trends	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
R^2	0.0227	0.0257	0.0260	0.0257	0.0335	0.0343			
N		25,039			8,177			8,169	

Robust standard errors in parentheses account for correlation between observations from the same site (1,363 clusters). All models include dummies for gender, race, and

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Table V: test scores and job spell duration

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TABLE VI

OLS AND IV ESTIMATES OF THE EFFECT OF JOB TESTING ON THE JOB SPELL DURATION OF HIRES: TESTING FOR DIFFERENTIAL IMPACTS BY RACE

(DEPENDENT VARIABLE: LENGTH OF COMPLETED EMPLOYMENT SPELL IN DAYS)

	(1)	(2)	(3)	(4)	(5)	(6)
	OLS estimates			2SLS estimates		
White × tested	13.8	19.7	23.2	12.3	17.0	20.4
	(5.0)	(4.6)	(4.8)	(5.7)	(5.2)	(5.6)
Black × tested	15.4	22.2	23.2	12.4	18.1	18.8
	(6.4)	(5.9)	(6.0)	(7.0)	(6.7)	(6.9)
Hispanic × tested	−1.2	7.0	12.8	−5.6	0.5	6.4
	(8.8)	(7.3)	(7.6)	(9.2)	(7.7)	(8.1)
Black	−44.5	−26.5	−25.8	−44.0	−26.2	−25.4
	(3.8)	(3.9)	(3.9)	(3.9)	(3.9)	(3.9)
Hispanic	−14.0	−8.2	−8.8	−13.1	−7.2	−7.8
	(5.5)	(4.8)	(4.9)	(5.6)	(4.9)	(4.9)
Male	−4.2	−2.0	−1.9	−4.2	−2.0	−1.9
	(2.4)	(2.4)	(2.4)	(2.4)	(2.4)	(2.4)
Site effects	No	Yes	Yes	No	Yes	Yes
State trends	No	No	Yes	No	No	Yes
H_0: Race interactions jointly equal	0.19	0.15	0.36	0.14	0.08	0.21
R^2	0.012	0.109	0.112			

$N = 33{,}266$. Robust standard errors in parentheses account for correlation between observations from the same site hired under each screening method (testing or no testing). All models include controls for month-year of hire. Sample includes workers hired January 1999 through May 2000 at 1,363 sites. Instrument for worker receiving employment test in columns (7)–(10) is an indicator variable equal to one if site has begun

Table VI: race-specific tenure effects

7.6 Figure V: Test Score Distribution of Hired Workers by Race + Table VII: Estimates of the Effect of Job Testing on Hiring Odds by Race

Read:

- Figure V shows that hired workers from different racial groups have similar score distributions.
- The figure explains why the score gap among applicants need not translate into a large score gap among hires.
- Table VII estimates the effect of testing on hiring odds and hiring shares by race.
- Point estimates are small and statistically imprecise.
- The evidence does not indicate that testing reduced black or Hispanic hiring odds.

Detailed interpretation: Figure V provides a visual explanation of the no-disparate-impact result: because the firm screens from the top tail, the selected workers look more similar across race groups than the applicant pool does. Table VII formalizes the claim using fixed-effects and IV-style specifications for hiring composition.

Economic message: Testing did not measurably reduce minority hiring in this setting, even though minorities scored lower in the applicant pool.

onate share of black and Hispanic applicants reside. The contrast between Figures II and V suggests that disparate hiring impacts were a real possibility.

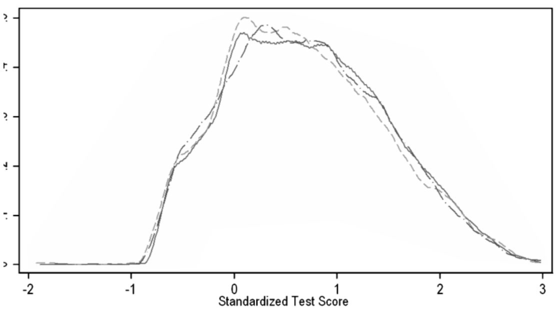

Figure V: score distribution of hired workers by race

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TABLE VII
ESTIMATES OF THE EFFECT OF JOB TESTING ON HIRING ODDS BY RACE (PANEL A) AND THE SHARE OF HIRES BY RACE (PANELS B AND C)
(DEPENDENT VARIABLE: EQUAL TO ONE (ZERO) IF HIRED WORKER IS (NOT) OF SPECIFIED RACE)

	(1)	(2)	(3)	(4)	(5)	(6)
	White		Black		Hispanic	
	Panel A. Hiring odds: 100 × fixed effects logit estimates					
Employment test	2.90	2.06	−2.35	−0.13	−2.48	−5.78
(logit coefficient)	(5.63)	(5.89)	(6.77)	(7.14)	(7.33)	(7.62)
State trends	No	Yes	No	Yes	No	Yes
N	30,921	23,957	26,982	26,982	22,453	22,453
	Panel B. Hiring shares: 100 × OLS estimates					
Employment test	0.41	0.24	−0.27	−0.04	−0.14	−0.21
(OLS coefficient)	(0.84)	(0.89)	(0.69)	(0.72)	(0.62)	(0.67)
State trends	No	Yes	No	Yes	No	Yes
N	33,924	33,924	33,924	33,924	33,924	33,924
	Panel C. Hiring shares: 100 × 2SLS estimates					
Employment test	0.78	0.69	−0.15	0.09	−0.63	−0.78
(2SLS coefficient)	(0.95)	(1.02)	(0.78)	(0.81)	(0.70)	(0.77)
State trends	No	Yes	No	Yes	No	Yes
N	33,924	33,924	33,924	33,924	33,924	33,924

Standard errors in parentheses. For OLS and IV models, robust standard errors in parentheses account for orrelations between observations from the same site. Sample includes workers hired January 1999 through Aay 2000. All models include controls for month-year of hire and site fixed effects. Fixed effects logit models

Table VII: hiring odds by race

7.7 Table VIII: Relationship Between Store ZIP Code Demographics and Race of Hires Before and After Applicant Testing

Read:

- Table VIII studies whether testing changes the relationship between local demographics and the race composition of hires.
- It compares hires before and after testing across race-specific columns.
- The results do not indicate that testing sharply changed stores' ability or tendency to hire workers from local minority populations.
- The table addresses the concern that applicant pools may shift after testing.

Detailed interpretation: Since pre-testing application flows by race are not fully observed, Table VIII is an important robustness check. It asks whether stores in minority areas show different post-testing hiring changes. The estimates do not point to a strong compositional break caused by testing.

Economic message: The minority-hiring result does not appear to be driven by large post-testing changes in local demographic sorting.

TABLE VIII
THE RELATIONSHIP BETWEEN STORE ZIP CODE DEMOGRAPHICS AND RACE OF HIRES BEFORE AND AFTER USE OF APPLICANT TESTING
(DEPENDENT VARIABLE: AN INDICATOR VARIABLE EQUAL TO 100 IF WORKER IS OF GIVEN RACE)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	White				Black				Hispanic			
	Not tested	Tested	All	All	Not tested	Tested	All	All	Not tested	Tested	All	All
Panel A: Race of hires and minority share in store ZIP code												
Share nonwhite	−87.4	−86.1	−87.6		56.5	56.7	56.5		30.9	29.4	31.2	
in ZIP code	(2.3)	(3.4)	(2.2)		(3.5)	(4.9)	(3.3)		(3.0)	(4.4)	(2.9)	
Share nonwhite			1.3	−0.3			1.1	1.3			−2.4	−1.1
× tested			(3.3)	(1.8)			(4.9)	(1.7)			(4.5)	(1.6)
Site effects	No	No	No	Yes	No	No	No	Yes	No	No	No	Yes
State effects	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No
R^2	0.232	0.253	0.236	0.353	0.169	0.197	0.174	0.356	0.130	0.110	0.124	0.296
N	25,561	8,363	33,924	33,924	25,561	8,363	33,924	33,924	25,561	8,363	33,924	33,924
Panel B: Race of hires and log median income in store ZIP code												
Log median income	32.0	39.5	32.2		−20.0	−23.0	−20.0		−12.1	−16.5	−12.3	
in ZIP code	(2.5)	(3.1)	(2.4)		(2.5)	(3.2)	(2.4)		(1.6)	(2.5)	(1.6)	
Log median income			5.9	0.6			−3.0	−0.4			−2.8	−0.3
× tested			(3.8)	(1.6)			(3.7)	(1.4)			(2.8)	(1.2)
Site effects	No	No	No	Yes	No	No	No	Yes	No	No	No	Yes
State effects	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No
R^2	0.117	0.155	0.125	0.353	0.099	0.129	0.104	0.356	0.102	0.095	0.099	0.296
N	25,561	8,363	33,924	33,924	25,561	8,363	33,924	33,924	25,561	8,363	33,924	33,924

Robust standard errors in parentheses account for correlations between observations from the same site (pre or post use of employment testing in models where both included). Sample includes workers hired January 1999 through May 2000. All models include controls for month-year of hire, and where indicated, 1,363 site fixed effects or state fixed effects

Table VIII: local demographics and hire race before/after testing

7.8 Table IX: The Impact of Job Testing under Six Bias Scenarios

Read:

- Table IX compares simulated productivity and hiring results under six combinations of test bias and interview bias.
- Scenarios where tests are unbiased and interviews are not strongly minority-disfavoring fit the observed data better.
- Scenarios where tests are majority-favoring imply much larger minority hiring losses than observed.
- The observed pattern combines large productivity gains with no large black-white hiring decline.
- The simulation therefore supports the interpretation that testing mainly improved precision rather than adding majority-favoring bias.

Detailed interpretation: Table IX is the paper's synthesis. It maps the conceptual framework into quantitative predictions. The central empirical pattern is hard to reconcile with a strongly biased test: a biased test should reduce minority hiring more visibly. The observed data look more consistent with a test that is relatively unbiased compared with prior interviews.

Economic message: The best-fitting scenario is not an equality-efficiency trade-off; it is an efficiency gain through better, relatively unbiased screening.

TABLE IX
THE IMPACT OF JOB TESTING ON HIRING AND JOB SPELL DURATIONS OF WHITE AND BLACK APPLICANTS UNDER SIX BIAS SCENARIOS: COMPARING SIMULATION RESULTS WITH OBSERVED OUTCOMES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Simulation Results						
Avg. ability	$W > B$	$W > B$	$W > B$	$W = B$	$W = B$	$W = B$	
Interview bias	Neutral	Favors W	Favors B	Neutral	Favors W	Favors B	
Test bias	Neutral	Neutral	Neutral	Favors W	Favors W	Favors W	Observed
A. Productivity: job spell durations in days							
Initial tenure	52.0	30.1	80.7	−13.2	−41.9	15.6	44.9
gap: $W - B$	(5.1)	(5.9)	(5.0)	(4.9)	(5.1)	(4.5)	(3.9)
ΔW tenure	18.6	20.4	16.8	16.8	18.6	16.0	23.2
	(1.2)	(1.1)	(1.3)	(1.3)	(1.2)	(1.3)	(4.8)
ΔB tenure	19.9	19.7	23.1	23.2	20.0	27.3	23.2
	(2.7)	(3.2)	(2.3)	(2.3)	(2.7)	(2.1)	(6.0)
$\Delta W - \Delta B$	−1.4	0.7	−6.3	−6.4	−1.4	−11.3	0.0
tenure	(3.0)	(3.4)	(2.7)	(2.7)	(3.0)	(2.6)	(6.2)
$\chi^2(3)$ rows 1, 2, 3	2.4	5.1	34.0	88.1	185.5	26.6	
P-value	.50	.17	.00	.00	.00	.00	

Table IX: simulation results, part 1

TABLE IX
(CONTINUED)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Simulation Results						
Avg. ability	$W > B$	$W > B$	$W > B$	$W = B$	$W = B$	$W = B$	
Interview bias	Neutral	Favors W	Favors B	Neutral	Favors W	Favors B	
Test bias	Neutral	Neutral	Neutral	Favors W	Favors W	Favors W	Observed
B. Employment shares and log odds of hiring							
ΔW emp	−0.97	−2.38	0.86	0.86	−0.98	2.69	0.24
share × 100	(0.18)	(0.18)	(0.18)	(0.18)	(0.19)	(0.19)	(0.89)
ΔB emp	0.82	1.72	−0.53	−0.53	0.82	−1.88	−0.04
share × 100	(0.15)	(0.15)	(0.16)	(0.15)	(0.15)	(0.16)	(0.72)
$\Delta W - \Delta B$ emp	−1.79	−4.10	1.39	1.39	−1.79	4.57	0.28
share × 100	(0.31)	(0.30)	(0.31)	(0.30)	(0.31)	(0.32)	(1.42)
$\chi^2(2)$ rows 6, 7	3.4	14.9	1.0	1.0	3.4	15.0	
P-value	.33	.00	.79	.79	.33	.00	
C. Omnibus goodness of fit statistics for productivity and employment							
$\chi^2(5)$ rows 5, 9	5.8	20.0	35.0	89.2	188.9	41.6	
P-value	.33	.00	.00	.00	.00	.00	

One thousand replications of each of six scenarios (corresponding to columns (1)–(6)) using 189,067 applicant files. In columns (1)–(6), standard deviations of estimates from 1,000 simulations are in parentheses. In column (7), standard errors from empirical estimates are in parentheses. Point estimates and standard errors for results in column (7) are obtained from Tables I, V, and VII. χ^2 goodness of fit statistics are obtained by comparing simulation estimates in columns (1)–(6) to observed outcomes in column (7). See text for details of simulation procedure.

Table IX: simulation results, part 2

8 Detailed result map

- **Test scores differ by race.** Minority applicants have lower mean standardized test scores than white applicants.
- **Tests predict hiring.** The firm strongly uses test scores to select applicants.
- **Testing improves productivity.** Tested hires stay longer on the job.
- **Productivity gains are broad.** Black and white hires see similar tenure gains from testing.
- **Minority hiring does not fall.** Fixed-effects and IV evidence shows no measurable negative effect on minority hiring shares.
- **Simulation supports precision-improvement interpretation.** The observed facts are most consistent with testing improving precision without adding group bias relative to interviews.

9 Short summary

- The paper studies a national retail firm's rollout of computerized job testing.
- The core question is whether testing creates an equality-efficiency trade-off.
- The conceptual framework shows that lower minority test scores do not automatically imply minority hiring losses.
- Testing can raise screening precision without changing beliefs about average group productivity.
- The data include nearly 190,000 applicants and over 33,000 hires.
- Minority applicants have lower average test scores, but minority hiring does not decline after testing.
- Testing increases job tenure, the main productivity proxy.
- Productivity gains are similar for black and white workers.
- Simulation evidence supports the view that testing was not majority-favoring relative to the previous interview process.
- The broader lesson is that the disparate impact of testing depends on relative bias between old and new screening methods.

10 Conclusion

10.1 Core conclusion

Autor and Scarborough show that job testing can improve worker selection without necessarily reducing minority hiring. In their retail setting, computerized testing raises worker tenure substantially, but has no measurable adverse effect on black or Hispanic hiring.

10.2 Economic intuition

The key intuition is that firms already screen applicants before formal testing. If informal interviews and formal tests are both noisy but unbiased predictors of productivity, replacing interviews with a more precise test improves selection mostly within groups rather than by shifting hiring sharply across groups. Disparate test scores among applicants matter, but they do not by themselves determine the employment effect of testing.

10.3 Incremental contribution to the literature

The paper's incremental contribution is to combine a clear theoretical distinction between precision and bias with unusually rich firm-level evidence. It contributes to labor economics by showing that personnel technologies can improve match quality. It contributes to discrimination research by showing that adverse impact cannot be inferred from group score differences alone. It contributes to policy debates by showing how Title VII-style concerns should evaluate the new screen relative to the screen it replaces.

10.4 Policy implications

The paper does not imply that all job tests are harmless. Instead, it implies that regulators and firms should ask three questions: whether the test predicts productivity, whether it is biased relative to the prior screen, and whether the prior screen was itself biased. A valid and relatively unbiased test may reduce subjective discretion and improve equality even when minority applicants score lower on average.

10.5 Limitations

The evidence comes from one retail firm and one class of hourly jobs. Productivity is proxied by tenure, not direct output or customer service quality. Application flows by race are not fully observed before testing, so some assumptions are needed in the minority-hiring analysis. The results should therefore be interpreted as strong evidence in this setting rather than proof about all employment tests.

10.6 Takeaway

The central takeaway is that the equality-efficiency trade-off in job testing is not mechanical. Testing can raise productivity and preserve minority opportunity when it improves screening precision without adding negative group bias relative to existing hiring methods.